

# Data Cleaning & Visualization Project - Python Code

Sample solution for Task 1: Data Cleaning & Visualization Project

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("data.csv")

# Display first 5 rows
print(df.head())

# Check missing values
print(df.isnull().sum())

# Remove duplicate rows
df = df.drop_duplicates()

# Fill missing numeric values with mean
for col in df.select_dtypes(include='number').columns:
    df[col] = df[col].fillna(df[col].mean())

# Basic statistics
print(df.describe())

# Visualization 1: Histogram
df.hist(figsize=(10, 6))
plt.tight_layout()
plt.show()

# Visualization 2: Correlation Heatmap
corr = df.corr(numeric_only=True)
plt.figure(figsize=(8, 6))
plt.imshow(corr)
plt.colorbar()
plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)
plt.yticks(range(len(corr.columns)), corr.columns)
plt.title("Correlation Matrix")
plt.show()

# Save cleaned dataset
df.to_csv("cleaned_data.csv", index=False)

print("Data cleaning and visualization completed successfully!")
```